Fuel Data Management System – C 12 T - I/O Interfaces.

Over REST API.

Specification.

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- C O N F I D E N T I A L –

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Overview

Requirements

Infrastructure.

The system should be hosted in a cloud infrastructure (server) and accessible via web-interface from any smart device with web-browser. The web-page format should be optimized for smart-phone and normal computer.

User interface.

The target users of the system would mostly use it on desktop/laptop computers but also on high quality mobile devices. As such the requirement is that the cloud application is mobile friendly and fully usable on a mobile device.

The interface should be light but with very strong visualization capabilities allowing filtering and sorting the results. The highest focus it to be set on visualization of the pumps/station metrics.

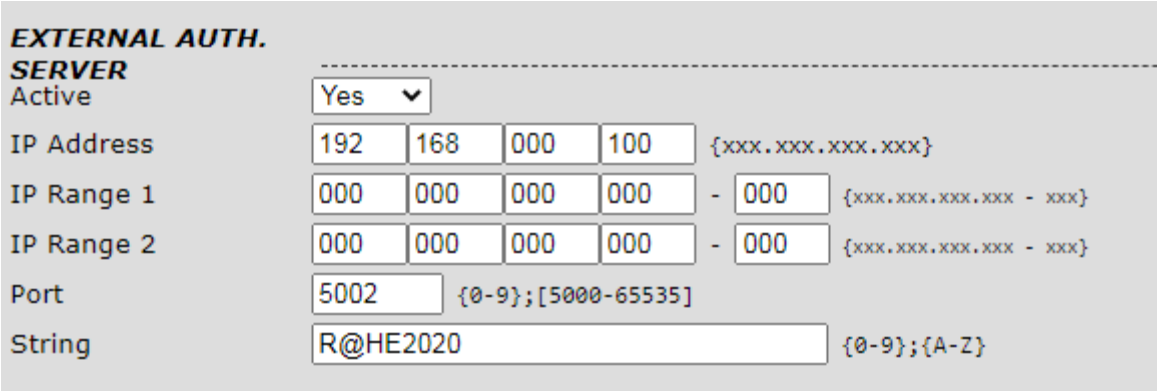
Data storage.

The data from the pump stations is stored in xml-file for each transaction. This xml file is then transmitted over the internet to an ftp-server. The cloud application needs to parse this data and store it in a database in order to aggregate all data and display it in the dashboards described below.

1. Transaction management using external host-server

1.1 Preconditions:

- FDC – FW version 3.49 (or higher)
- The "External Authorization Server" (EAS) is in the same network as FDC
- The IP address of the EAS is allowed by the FDC (see below picture)
- The PORT used for the transaction management is PORT+1 from the defined port at "External Authorization Server" configuration. For example, if the specified port for the external authorization is 5002 then the transaction management port will be 5003.



- Each message contains a SIGNATURE tab. The rules for calculating the signature are described below and are the same as for the "External Authorization"

1.2 Calculating the SIGNATURE of the messages:

The content closed between the tabs: <SIGNATURE> is SHA1 calculation of the content b/w <REQ> </REQ> tabs (including them), plus the secret word (password) added immediately after </REQ> (with no space in between).

Note: the SHA1 calculation can be verified with the online calculator given below:

https://md5calc.com/hash/sha1

The value should be exactly the same as with the result generated by the above tool!
Note that all spaces and special symbols like /n matter and will change the signature!

1.3 Host communication rules:

- For communicating with FDC is used REST API's POST (posting) request from the host.
- The requested device is always supposed to return an answer in order to close the dialog and consider the information as successfully transferred.

1.4 Header format of the POST message (from Host to FDC):

Custom Header Keys:

- Content-Type: "Application/xml"
- USN-Code: "1 – 999999" (Unique Station Number) – see Note1 below
- Operation-Code: "1"

OPERATION CODES:

1- TRANSACTION.
2- PRODUCTS and PRICES (explained in the next chapters)
3- DAY CLOSE (explained in the next chapters)
4- ATG DATA (explained in the next chapters)
5- CSR DATA (explained in the next chapters)

Note 1:

The USN-Code should be same as the FDC number given in the system configuration.

Note 2:

The header should also contain the mandatory key 'Content-Length' for message's body length in bytes.

1.5 Common parameters of the XML-message transferred between the devices.

- <CMD_CODE>10</CMD_CODE> - command code (this is a mandatory parameter for each message);

COMMAND CODES:

CMD_CODE = 10 - HOST REQUEST FOR TRANSACTION
CMD_CODE = 20 - MODE CHANGE

- <CMD_NAME>TRN_REQ</CMD_NAME> - command name (command description), this tab is optional, it is used for logging and more explanation purposes and can be left empty;

- <TOKEN>12345</TOKEN> - a string of max 16 digits (only decimal digits are allowed). It is used to identify that the communication sequence is the same between the host and the FDC. Recommended to have each token with unique number;

- RESPONSE CODES – these are the codes returned by the FDC after initiated request

RESPONSE CODES

201 – OK / SUCCESS

205 – NO TRANSACTION AVAILABLE

- ERROR CODES

These are the codes returned after an error, wrong format, wrong signature, etc. to inform the sending device that the information was NOT transferred successfully.

ERROR MESSAGES:

1. 206 – TRANSACTION MODE ERROR
2. 251 – SIGNATURE ERROR
3. 253 – TOKEN ERROR
4. 255 – BAD XML FORMAT

1.6 Messages from Host to FDC:

- **MODE CHANGE** (20) – Changing the way/mode of how the transactions are transferred from FDC to the host.

MODES:

1 - ON_DEMAND – transfer on host request
2 - UNSOLICITED – unsolicited request from FDC
0 - MODE OFF

This command is allowed in all modes as long as the other communication rules are fulfilled. The host should receive answer from FDC after a successful transmission of this command!

- **REQUEST FOR TRANSACTION** (10)

Requesting the oldest transaction from FDC buffer.

- This command is allowed only in mode „1" - ON_DEMAND. The other modes will ignore it.
- The host should receive answer from FDC after a successful transmission of this command!

1.7 Messages from FDC to Host:

- **POST A TRANSACTION** – Sends the oldest unsent transaction from the buffer in UNSOLICITED mode.

This command is allowed only in mode „UNSOLICITED ".
The host should respond to FDC after a successful transmission of the transaction.

1.8 EXAMPLES:

- <u>MODE CHANGE / SET TO "ON-DEMAND" (from HOST to FDC)</u>

  - It is allowed in all modes even the current mode is "OFF"!

Header (Custom tabs):

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
  <REQ>
    <CMD_CODE>20</CMD_CODE>
    <CMD_NAME>MODE_CNG</CMD_NAME>
    <MODE>1</MODE>
    <TOKEN>12345</TOKEN>
  </REQ>
```

```
  <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

FDC REPLY (when the request is successfully accepted):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
 <TABLE VERSION="1.0">
  <ANS
   RESP_CODE="201"
   RESP_MSG="ON_DEMAND"
   TOKEN="12345"
  />
 </TABLE>
 <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

FDC REPLY (when error is detected):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
  <TABLE VERSION="1.0">
    <ANS
      RESP_CODE="251"
      RESP_MSG="SIGNATURE_ERR"
      TOKEN="12345"
    />
  </TABLE>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

- HOST REQUEST FOR TRANSACTION (from HOST to FDC) –when FDC is already set to "ON DEMAND" mode

Header:

```
Content-Type: Application/xml
USN-Code: 123
Operation: "1"
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
    <REQ>
```

```
      <CMD_CODE>10</CMD_CODE>
      <CMD_NAME>TRN_REQ</CMD_NAME>
      <TOKEN>12345</TOKEN>
   </REQ>
   <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

FDC REPLY (when successfully accepted and TRN available):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
  <TABLE VERSION="1.0">
        <ANS
              RESP_CODE="201"
              RESP_MSG="SUCCESS"
              TOKEN="12345"
        />
        <TRN
              AMO="30000.0"
              EFD_ID="182AC9368989"
              FDC_DATE="2021-03-03"
              FDC_NAME="10TZ100449"
              FDC_NUM="100253410"
              FDC_PROD="0"
              FDC_PROD_NAME="UNLEADED"
              FDC_SAVE_NUM="368989"
              FDC_TANK=""
              FDC_TIME="21:17:53"
              FP="0"
              NOZ="0"
              PRICE="1930"
              PUMP_ADDR="0"
```

```
            RDG_DATE="2021-03-03"
            RDG_ID="0"
            RDG_INDEX="0"
            RDG_PROD="0"
            RDG_SAVE_NUM="1066"
            RDG_TIME="21:17:53"
            REG_ID="TZ0100551361"
            ROUND_TYPE="0"
            VOL="15.54"
        />
        <RFID_CARD
            CARD_TYPE="6"
            CUST_CONTACT=""
            CUST_ID="0"
            CUST_IDTYPE="6"
            CUST_NAME=""
            DISCOUNT="0"
            DISCOUNT_TYPE="1"
            NUM="0"
            NUM_10="0"
            PAY_METHOD="0"
            PRODUCT_ENABLED="0"
            USED="0"
        />
        <DISCOUNT
            AMO_DISCOUNT="0.0"
            AMO_NEW="30000.0"
            AMO_ORIGIN="30000.0"
            DISCOUNT_TYPE="1"
            PRICE_DISCOUNT="0"
            PRICE_NEW="1930"
            PRICE_ORIGIN="1930"
            VOL_ORIGIN="15.54"
        />
       <CUST_DATA USED="0">
  </TABLE>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

HOST REPLY (when successfully accepted):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
  <REQ>
    <CMD_CODE>201</CMD_CODE>
    <CMD_NAME>SUCCESS</CMD_NAME>
    <TOKEN>12345</TOKEN>
  </REQ>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

HOST REPLY (when error occurred):

When the host detects an error in the received unsolicited transaction it simply does not respond. After a timeout occurs the FDC will send the same transaction again.
Note that it is assumed that the FDC always transfers the data in the right format and proper signature which means that an error can occur only in the transportation layer between the two devices.

FDC REPLY (when successfully accepted but TRN is NOT available):

Header:

```
Content-Type: Application/xml
USN-Code: 1
```

Operation: 1

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
  <TABLE VERSION="1.0">
    <ANS
      RESP_CODE="205"
      RESP_MSG="NO TRN AVAILABLE"
      TOKEN="12345"
    />
    <TRN />
    <RFID_CARD />
    <DISCOUNT />
  </TABLE>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

<u>FDC REPLY (when error is detected):</u>

Header:

Content-Type: Application/xml
USN-Code: 1
Operation: 1

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
  <TABLE VERSION="1.0">
    <ANS
      RESP_CODE="251"
      RESP_MSG="SIGNATURE_ERR"
      TOKEN="12345"
    />
  </TABLE>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

- FDC POST A TRANSACTION UNSOLICITED *(from FDC to HOST)*

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
```

```
<TABLE VERSION="1.0">
    <ANS
        RESP_CODE="30"
        RESP_MSG="UNSOL_TRN"
        TOKEN="12345"
    />
    <TRN
        AMO="30000.0"
        EFD_ID="182AC9368989"
        FDC_DATE="2021-03-03"
        FDC_NAME="10TZ100449"
        FDC_NUM="100253410"
        FDC_PROD="0"
        FDC_PROD_NAME="UNLEADED"
        FDC_SAVE_NUM="368989"
        FDC_TANK=""
        FDC_TIME="21:17:53"
        FP="0"
        NOZ="0"
        PRICE="1930"
        PUMP_ADDR="0"
        RDG_DATE="2021-03-03"
        RDG_ID="0"
        RDG_INDEX="0"
        RDG_PROD="0"
        RDG_SAVE_NUM="1066"
        RDG_TIME="21:17:53"
        REG_ID="TZ0100551361"
        ROUND_TYPE="0"
        VOL="15.54"
    />
    <RFID_CARD
        CARD_TYPE="6"
        CUST_CONTACT=""
        CUST_ID="0"
        CUST_IDTYPE="6"
        CUST_NAME=""
        DISCOUNT="0"
        DISCOUNT_TYPE="1"
```

```
                NUM="0"
                NUM_10="0"
                PAY_METHOD="0"
                PRODUCT_ENABLED="0"
                USED="0"
            />
            <DISCOUNT
                AMO_DISCOUNT="0.0"
                AMO_NEW="30000.0"
                AMO_ORIGIN="30000.0"
                DISCOUNT_TYPE="1"
                PRICE_DISCOUNT="0"
                PRICE_NEW="1930"
                PRICE_ORIGIN="1930"
                VOL_ORIGIN="15.54"
            />
           <CUST_DATA USED="0">

  </TABLE>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```


HOST REPLY (when successfully accepted):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:


```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
   <REQ>
      <CMD_CODE>201</CMD_CODE>
      <CMD_NAME>SUCCESS</CMD_NAME>
      <TOKEN>12345</TOKEN>
   </REQ>
   <SIGNATURE>ABCDF12345</SIGNATURE>
```

```
</HOST_REQ>
```

HOST REPLY (when error occurred):

When the host detects an error in the received unsolicited transaction it simply does not respond. After a timeout occurs the FDC will send the same transaction again.
Note that it is assumed that the FDC always transfers the data in the right format and proper signature which means that an error can occur only in the transportation layer between the two devices.

2. Read and change product's prices and parameters

2.1 Preconditions:

- FDC – FW version 3.49 (or higher)
- The "External Authorization Server" (EAS) is in the same network as FDC
- The IP address of the EAS is allowed by the FDC
- DSB mode: POS dependent
- The PORT used for the transaction management is PORT+1 from the defined port at "External Authorization Server" configuration. For example, if the specified port for the external authorization is 5002 then the transaction management port will be 5003.
- Each message contains a SIGNATURE tab. The rules for calculating the signature are described below and are the same as for the "External Authorization"

2.2 Calculating the SIGNATURE of the messages:

The content closed between the tabs: <SIGNATURE> is SHA1 calculation of the content b/w <REQ> </REQ> tabs (including them), plus the secret word (password) added immediately after </REQ> (with no space in between).

Note: the SHA1 calculation can be verified with the online calculator given below:

https://md5calc.com/hash/sha1

The value should be exactly the same as with the result generated by the above tool!
Note that all spaces and special symbols like /n matter and will change the signature!

2.3 Host communication rules:

- For communicating with FDC is used REST API's POST (posting) request from the host.
- The requested device is always supposed to return an answer in order to close the dialog and consider the information as successfully transferred.

2.4 Header format of the POST message (from Host to FDC):

Format of the POST message (from Host to FDC where FDC is server):

MANDATORY HEADER KEYS:

- Content-Type: "Application/xml"
- USN-Code: "1 – 999999" (Unique Station Number) – see Note1 below
- Operation-code: 2 or 3

Note1: the USN-Code should be same as the FDC number given in the system configuration.
Note2: the header should also contain the standard key 'Content-Length' for message's body length in bytes.

OPERATION CODES:

2- PRODUCTS and PRICES
3- DAY CLOSE

COMMAND CODES:
55 - HOST REQUESTS – READ PRICES AND PRODUCTS
66 - HOST WRITES – CHANGE/WRITE PRICES AND PRODUCTS
77 - HOST WRITES – DAY CLOSE

2.5 EXAMPLES:

- HOST REQUEST:

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
<REQ>
        <CMD_CODE>55</CMD_CODE>
        <CMD_NAME>PROD and PRICE REQ</CMD_NAME>
        <TOKEN>12345</TOKEN>
</REQ>
<SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

FDC RESPONSE (when request is OK):

```
<FDC_RESP>
 <TABLE VERSION="1.0">
  <ANS
   RESP_CODE="201"
   RESP_MSG="REQ OK"
   TOKEN="12345"
  />
  <PRODUCT ID="1">
   NAME="Product_1"
   PRICE="10.99"
  />
  <PRODUCT ID="2">
   NAME="Product_2"
   PRICE="22.99"
  />
```

```
  <PRODUCT ID="3">
    NAME="Product_3"
    PRICE="33.99"
  />
 </TABLE>
 <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

Note:
- Only active (activated) products are included in the response. Inactive ones are not included and sent.

FDC RESPONSE (when error in the request is detected):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
<TABLE VERSION="1.0">
<ANS
RESP_CODE="251"
RESP_MSG="SIGNATURE_ERR"
TOKEN="12345"
/>
</TABLE>
<SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

- CHANGING PRODUCT AND/OR PRICE PARAMETERS (Host Write Request):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 1
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
<REQ>
        <CMD_CODE>66</CMD_CODE>
        <CMD_NAME>PROD and PRICE CHANGE</CMD_NAME>
        <TOKEN>12345</TOKEN>
        <PRODUCT_ID>1</PRODUCT_ID>
        <PRODUCT_NAME>DIESEL</PRODUCT_NAME>
        <PRODUCT_PRICE>10.99</PRODUCT_PRICE>
        <PRODUCT_ACTIVE>YES</PRODUCT_ACTIVE>
        <PRODUCT_ENABLE>YES</PRODUCT_ENABLE>
 </REQ>
        <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

Notes:

- Only the products whose one or more parameters will be changed are included in the request, the remaining, unmentioned products in the FDC are kept unchanged regardless of whether they are active or not.
- Each request is for a separate product
- When PRODUCT_ACTIVE = No, then the product will disappear from the reports, make sure that there are no nozzles attached to it.
- When PRODUCT_ENABLE = No, then the product is visible but the nozzles attached to it will not receive authorization.

FDC RESPONSE (when OK):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
<TABLE VERSION="1.0">
<ANS
RESP_CODE="201"
RESP_MSG="REQ OK"
TOKEN="12345"
/>
</TABLE>
<SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

FDC RESPONSE (when error in the request is detected):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
<TABLE VERSION="1.0">
<ANS
RESP_CODE="251"
RESP_MSG="SIGNATURE_ERR"
TOKEN="12345"
/>
</TABLE>
<SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

- DAY CLOSE (Host Write Request):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
  <REQ>
    <CMD_CODE>77</CMD_CODE>
```

```
  <CMD_NAME>DAY CLOSE</CMD_NAME>
  <CLOSE_IMMEDIATE>YES</CLOSE_IMMEDIATE>
  <CLOSE_TIME>18:02:00</CLOSE_TIME>
  <TOKEN>12345</TOKEN>
 </REQ>
 <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

Note:

- If CLOSE_IMMEDIATE="YES" then CLOSE_TIME does not matter. I.e. the day is closed immediately after receiving the message. Conversely, the day will close depending on "CLOSE_TIME" value.
- The time is valid only within the current day.

FDC RESPONSE (when OK):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
<TABLE VERSION="1.0">
  <ANS
        RESP_CODE="201"
        RESP_MSG="REQ OK"
        TOKEN="12345"
  />
</TABLE>
<SIGNATURE>ABCDF12345</SIGNATURE>
```

```
</FDC_RESP>
```

ERROR CODES:

201 – SUCCESS / ACCEPTED
207 - REST_API_ACK_CODE_PROD_DATA_ERROR
251 – SIGNATURE ERROR
255 – BAD XML FORMAT
256 – BAD HEADER FORMAT
258 - PUMP NOT READY
260 – DSB IS OFFLINE

3. External Authorization

3.1 Preconditions:

- FDC – FW version 3.49 (or higher)
- The "External Authorization Server" (EAS) is in the same network as FDC
- The IP address of the EAS is allowed by the FDC
- DSB mode: POS dependent
- The PORT used for the transaction management is from the defined port at "External Authorization Server" configuration from FDC.
- Each message contains a SIGNATURE tab. The rules for calculating the signature are same as explained in the previous chapters.

3.2 Format of the POST message (from Host to Comforter):

Mandatory Header Keys :

- Content-Type: "Application/xml"
- USN-Code: "1 – 999999" (Unique Station Number) – see Note1 below
- Operation: "Authorize"

Note1: the USN-Code should be same as the FDC number given in the system configuration.
Note2: the header should also contain the standard key 'Content-Length' for message's body length in bytes.

Body:

```
<?xml version="1.0" encoding="utf-8"?>
<FDCMS>
<AUTH_DATA>
        <PUMP>3</PUMP>
        <FP>1</FP>
        <AUTH>TRUE</AUTH>
        <PROD>2</PROD>
        <PRESET_VOLUME>0.00</PRESET_VOLUME>
        <PRESET_AMOUNT>2000</PRESET_AMOUNT>
        <CUSTNAME>TECHOLOGY Ltd.</CUSTNAME>
        <CUSTIDTYPE>1</CUSTIDTYPE>
```

```
        <CUSTID>12345678</CUSTID>
        <MOBILENUM>25588776655</MOBILENUM>
        <DISC_VALUE>10</DISC_VALUE>
        <DISC_TYPE>VALUE</DISC_TYPE>
        <TOKEN>123456</TOKEN>
</AUTH_DATA>
<FDCSIGNATURE>484000e79308e7e827359bc90069555115130860</FDCSIG
NATURE>
</FDCMS>
```


WHERE:

<PUMP> - the number of the RDG/DSB in the FDC, assigned to the particular fuel-pump
<FP> - the filling point number within the current RDG/DSB
<AUTH> - TRUE: activating the authorization, FALSE: canceling an active authorization
<PROD> - product number to be used, when '0' then all products can be used
<PRESET_VOLUME> - preseted volume in Liters (or Galons)
<PRESET_AMOUNT> - preseted value in the local currency

<CUSTNAME> - (optional) Customer or company name, if provided it will be assigned to the transaction data.
<CUSTIDTYPE> - (optional) Customer ID type from 1 to 6. If provided it will be assigned to the transaction data.

ID TYPE DESCRIPTION:

1=TIN (Tax identification number)
2=Driving License
3=Voters Number
4=Passport
5=NID (National Identity)
6=NIL (If there is no ID)

- <CUSTID> - (optional) Customer ID – a unique number identifying the particular customer. If provided it will be assigned to the transaction data.

- <MOBILENUM> - (optional) Customer phone number. If provided it will be assigned to the transaction data.

- <DISC_VALUE> - (optional) Discount value either in percent or in money per liter. Note: The value should be negative if negative discount is required. Example: <DISC_VALUE>-10</DISC_VALUE>

- <DISC_TYPE> - (optional) Type of the discount:

  - PERCENT = Percent discount
  - VALUE= Money discount

- <TOKEN> - (optional) Token value from 0 – 65535 that identifies the transaction delivered via the particular preset. This token will be included in the transaction data initiated by the given preset.

<FDCSIGNATURE> - the SHA1 calculation of the content b/w <AUTH_DATA> </AUTH_DATA> (including) plus a secret word (password) added immediately after
</AUTH_DATA> with no space in between.

FDC REPLY (on POST request from the HOST)

In case of proper sending of the POST request to the FDC one of the following answers is provided:

Header:

```
Content-Type: Application/xml
USN-Code: Unique Station Number from 1 to 999999
```

Body (WHEN OK):

```
<?xml version="1.0" encoding="UTF-8"?>
<FDCMS>
  <FDCACK>
    <DATE>2021-03-01</DATE>
```

```
      <TIME>09:38:42</TIME>
      <ACKCODE>0</ACKCODE>
      <ACKMSG>Success</ACKMSG>
   </FDCACK>
<FDCSIGNATURE>B7D5809F3CA32DDCA4FB084D5ED9D87DDB65</FDCSIG
NATURE>
</FDCMS>
```


Body (when the massage is rejected due to a format or signature error):


```
<?xml version="1.0" encoding="UTF-8"?>
<FDCMS>
   <FDCACK>
      <DATE>2021-03-01</DATE>
      <TIME>10:17:12</TIME>
      <ACKCODE>251</ACKCODE>
      <ACKMSG>Error</ACKMSG>
   </FDCACK>
<FDCSIGNATURE>B08A64EDEAB9EB8EA6455FBFDE7712702E06D</FDCSIGN
ATURE>
</FDCMS>
```


<u>RESPPONSE CODES:</u>

1. 0 – OK / SUCCESS
2. 251 – SIGNATURE ERROR
3. 255 – BAD XML FORMAT
4. 256 – BAD HEADER FORMAT
5. 258 - PUMP NOT READY
6. 260 – DSB IS OFFLINE

4. ATG Data reception

OPERATION CODES:

1- TRANSACTION
2- PRODUCTS and PRICES
3- DAY CLOSE
4 - ATG DATA

COMMAND CODES:

1. 10 - HOST REQUEST FOR TRANSACTION
2. 20 - HOST WRITES – MODE CHANGE
3. 55 - HOST REQUESTS – READ PRICES AND PRODUCTS
4. 66 - HOST WRITES – CHANGE/WRITE PRICES AND PRODUCTS
5. 77 - HOST WRITES – DAY CLOSE
6. 30 – HOST REQUEST - TANK DATA
7. 35 – HOST REQUEST - ATG DELIVERY

4.1 Tank-level data – Host Request:

Header (Custom tabs):

```
Content-Type: Application/xml
USN-Code: 1
Operation: 4
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
  <REQ>
    <CMD_CODE>30</CMD_CODE>
    <CMD_NAME>ATG_DATA_REQ</CMD_NAME>
    <TOKEN>12345</TOKEN>
  </REQ>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

FDC REPLY (when the request is successfully accepted):

Header:

Content-Type: Application/xml USN-Code: 1 Operation: 4

Body:


```xml
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
	<TABLE VERSION="1.0">
		<ANS
		RESP_CODE="201"
		RESP_MSG="SUCCESS"
		TOKEN="12345"
		/>
		<VR_INV_AL_INFO
			DATE="2023-11-06"
			TIME="23:10:22"
		/>
		<TANK_DATA>
			<TANK
				ID="0"
			PROD_NUM="0"
			VOL="1010.99"
			TC_VOL="1110.99"
			ULLAGE="111.99"
			HEIGHT="111.99"
			WATER="111.99"
			TEMP="+11.9"
			WATER_VOL="111.99"
			AL_MASK="0001"
			AL_NUM="1"
			/>
			<TANK
				ID="1"
```

```
                    PROD_NUM="1"
                    VOL="1010.99"
                    TC_VOL="1110.99"
                    ULLAGE="111.99"
                    HEIGHT="111.99"
                    WATER="111.99"
                    TEMP="+11.9"
                    WATER_VOL="111.99"
                    AL_MASK="0001"
                    AL_NUM="1"
                    />
                </TANK_DATA>
        </TABLE>
        <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```


4.2 ATG Deliveries Data – Host Request:

Header (Custom tabs):

```
Content-Type: Application/xml
USN-Code: 1
Operation: 4
```

Body:


```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
  <REQ>
    <CMD_CODE>35</CMD_CODE>
    <CMD_NAME>ATG_DEL_REQ</CMD_NAME>
    <TOKEN>12345</TOKEN>
  </REQ>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

FDC REPLY (when successfully accepted and new DELIVERY is available):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
 <TABLE VERSION="1.0">
        <ANS
              RESP_CODE="201"
              RESP_MSG="SUCCESS"
              TOKEN="12345"
        />
        <DELIVERY
              Whatever parameters available on the delivery
        />
 </TABLE>
 <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

HOST REPLY (when successfully accepted):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 4
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
   <REQ>
      <CMD_CODE>201</CMD_CODE>
      <CMD_NAME>SUCCESS</CMD_NAME>
      <TOKEN>12345</TOKEN>
   </REQ>
```

```
  <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

HOST REPLY (when error occurred):

When the host detects an error in the received unsolicited transaction it simply does not respond. After a timeout occurs the FDC will send the same transaction again.

FDC REPLY (when successfully accepted but new DELIVERY is NOT available):

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
 <TABLE VERSION="1.0">
  <ANS
   RESP_CODE="205"
   RESP_MSG="NO DEL AVAILABLE"
   TOKEN="12345"
  />
  <DELIVERY />
 </TABLE>
 <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```

RESPONSE CODES
201 – OK / SUCCESS
205 – NO DELIVERY AVAILABLE

ERROR MESSAGES:
251 – SIGNATURE ERROR
253 – TOKEN ERROR
255 – BAD XML FORMAT

5. CSR Data

OPERATION CODES:

1- TRANSACTION
2- PRODUCTS and PRICES
3- DAY CLOSE
4 - ATG DATA
5 - CSR DATA

COMMAND CODES:

1. 10 - HOST REQUEST FOR TRANSACTION
2. 20 - HOST WRITES – MODE CHANGE
3. 55 - HOST REQUESTS – READ PRICES AND PRODUCTS
4. 66 - HOST WRITES – CHANGE/WRITE PRICES AND PRODUCTS
5. 77 - HOST WRITES – DAY CLOSE
6. 30 – HOST REQUEST - TANK DATA
7. 35 – HOST REQUEST - ATG DELIVERY
8. 40 – HOST REQUEST - CSR DATA

5.1 CSR DATA - Host Request:

Header (Custom tabs):

```
Content-Type: Application/xml
USN-Code: 1
Operation: 5
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
  <REQ>
    <CMD_CODE>40</CMD_CODE>
```

```
    <CMD_NAME>CSR_DATA_REQ</CMD_NAME>
    <TOKEN>12345</TOKEN>
  </REQ>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

FDC REPLY (when the request is successfully accepted):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 5
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
        <TABLE VERSION="1.0">
                <ANS
                RESP_CODE="201"
                RESP_MSG="SUCCESS"
                TOKEN="12345"
                />
                <CSR>

                        The binary information in ASCII in the same format as
                        submitted by VFD-to COLLECT- at the same moment. (See
                        the note below)

                </CSR>
        </TABLE>
        <SIGNATURE>ABCDF12345</SIGNATURE>
```

</FDC_RESP>

Note:

At VFD **and** FDC1 **the number of** DSB/RDG is 12 while here it should be 16 (**from 0 to 15**).

HOST REPLY (when successfully accepted):

Header:

```
Content-Type: Application/xml
USN-Code: 1
Operation: 5
```

Body:

```
<?xml version="1.0" encoding="UTF-8"?>
<HOST_REQ>
   <REQ>
      <CMD_CODE>201</CMD_CODE>
      <CMD_NAME>SUCCESS</CMD_NAME>
      <TOKEN>12345</TOKEN>
   </REQ>
   <SIGNATURE>ABCDF12345</SIGNATURE>
</HOST_REQ>
```

HOST REPLY (when error occurred):

When the host detects an error in the received transaction it simply does not respond. FDC will send the same transaction again.

FDC REPLY (when sucessfully accepted but new CSR is NOT available):

Body:

```xml
<?xml version="1.0" encoding="UTF-8"?>
<FDC_RESP>
  <TABLE VERSION="1.0">
    <ANS
      RESP_CODE="205"
      RESP_MSG="NO NEW CSR AVAILABLE"
      TOKEN="12345"
    />
    <CSR></CSR>
  </TABLE>
  <SIGNATURE>ABCDF12345</SIGNATURE>
</FDC_RESP>
```


RESPONSE CODES.

1. 201 – OK / SUCCESS
2. 205 – NO CSR AVAILABLE

ERROR MESSAGES:

1. 207 – ACK CODE DATA ERROR
2. 251 – SIGNATURE ERROR
3. 252 – CMD CODE ERROR
4. 253 – TOKEN ERROR
5. 255 – BAD XML FORMAT